

WASP-36b

R-0986

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_180410 · created 2026-07-29T22:19:48+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458218.7151 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.134 +/- 0.048
Transit depth (Rp/Rs)^2	1.8 +/- 1.28 [%]
Orbital Inclination (inc)	80.24 +/- 2.88
Airmass coefficient 1 (a1)	1.02 +/- 0.49
Airmass coefficient 2 (a2)	0.001 +/- 0.098
Scatter in the residuals of the lightcurve fit is	49.17 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	3.0
Transit Duration (day)	0.039 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.134 ± 0.048 vs 0.1368 (deviation 0.06σ)
Duration fit vs published	0.039 d vs 0.0758 d (deviation 1.12σ)
Mid-time O–C	-5.4 min
Depth SNR	2.7
Residual scatter	49.17 %

Light curve

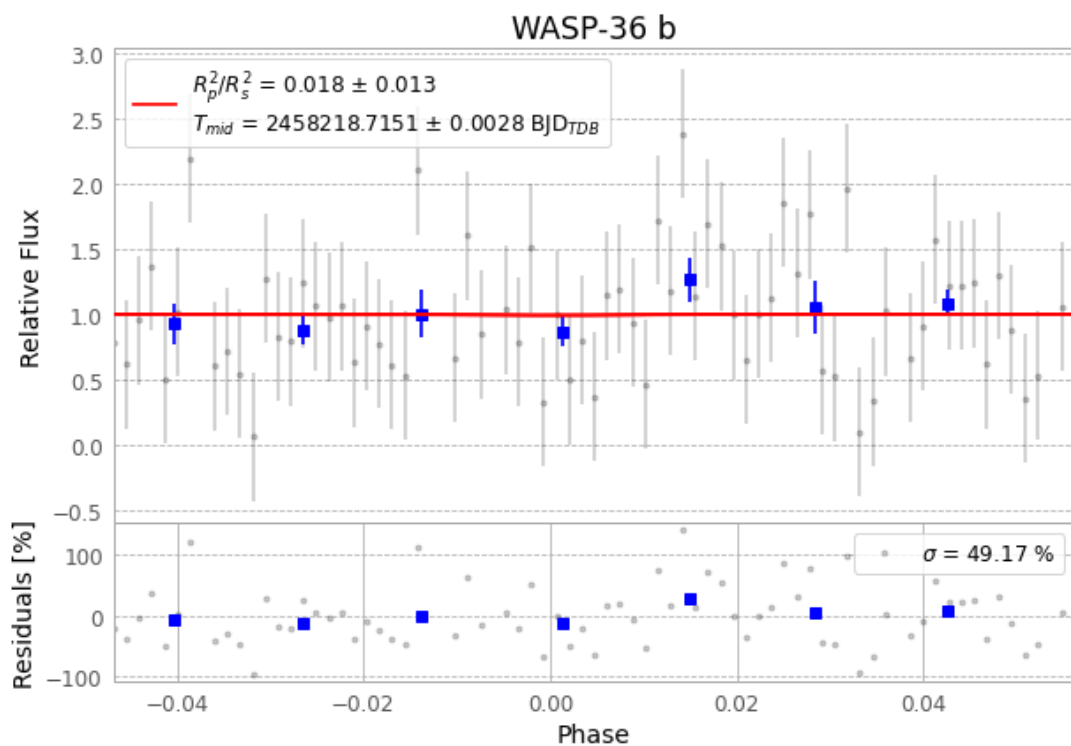


Figure 1 — FinalLightCurve_WASP-36 b_2018-04-09.png (SHA-256 222dd260bb98bea6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [410, 428], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 eb0cac5f73eb24e5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), WASP-36180410032112.FITS → WASP-36180410070912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-180410.log (a803169d8f5a...), FinalLightCurve_WASP-36 b_2018-04-09.png (222dd260bb98...), FinalLightCurve_WASP-36 b_2018-04-09.csv (0bee9db6149e...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 22:19Z.